

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0718

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Question 1

A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Rubric-Based Answer

Understanding of Problem & Constraints ()

The office is rural, wired high-speed Internet is unavailable, cloud applications and video conferencing need adequate bandwidth, headquarters communication must be secure, the budget is limited, operations must remain available, and the design must support future growth.

Problem Decomposition ()

The problem is divided into WAN selection, backup connectivity, bandwidth management, networking devices, security, reliability, cost control and scalability.

Application of Constraints in Solution ()

Use 4G/5G fixed wireless as primary WAN where coverage is reliable; use satellite as backup; use a dual-WAN/SD-WAN router; deploy firewall, managed switch, secure APs and site-to-site VPN; apply QoS to prioritize business and video traffic.

Logical Reasoning & Strategy ()

4G/5G provides better cost/performance for normal operation. Satellite is retained mainly for failover, reducing recurring cost. QoS protects video conferencing from non-critical traffic. VPN and firewall address secure HQ communication.

Solution Accuracy & Feasibility ()

The design is feasible for a remote office because it does not depend on wired broadband. Automatic WAN failover improves continuity, while managed networking equipment allows additional users and APs later.

Creativity & Innovation ()

A hybrid dual-WAN approach combines terrestrial wireless performance with satellite resilience instead of relying on one expensive connection.

Clarity of Presentation ()

The recommendation is summarized through constraints, design decisions, security measures and a justification table.

Constraint Application Summary

Constraint	Design Decision	Expected Benefit
Constraint	Design Decision	Expected Benefit
Bandwidth	4G/5G + QoS	Supports cloud services and video conferencing where coverage is good.
Reliability	Satellite backup + automatic failover	Maintains connectivity during primary-link failure.
Security	Firewall + IPsec VPN + VLANs	Protects branch-to-HQ and internal traffic.
Cost	Satellite used mainly for backup	Controls recurring connectivity expenditure.
Scalability	Managed switch + SD-WAN-ready gateway	Supports future users and additional WAN capacity.

Recommended Network Architecture

4G/5G Primary WAN → Dual-WAN/SD-WAN Router → Firewall → Managed Switch → Employee/Guest VLANs → Secure APs; IPsec VPN connects the branch securely to headquarters. Satellite remains available as backup.

Question 2

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Rubric-Based Answer

Understanding of Problem & Constraints ()

The network must serve more than 3,000 students across several building types, support high simultaneous-user density, minimize interference, provide secure access and operate with a limited AP budget.

Problem Decomposition ()

The problem is divided into site survey/AP placement, capacity planning, frequency selection, channel/interference management, authentication, segmentation and cost optimization.

Application of Constraints in Solution ()

Use centrally managed high-density Wi-Fi; place APs according to user density; use 5 GHz as the primary high-capacity band and 6 GHz where supported; use 2.4 GHz selectively; use RF/channel planning, transmit-power control, 802.1X/RADIUS and VLANs.

Logical Reasoning & Strategy ()

APs should be concentrated where users are dense rather than distributed only for maximum coverage. More efficient channel reuse and controlled power reduce interference. Enterprise authentication provides individual access control.

Solution Accuracy & Feasibility ()

A site-survey-driven design is more feasible than deploying APs uniformly. Central management allows monitoring and later expansion. Wired PoE uplinks provide stable AP backhaul.

Creativity & Innovation ()

Capacity-based AP placement and band steering/channel reuse make better use of a limited AP budget.

Clarity of Presentation ()

The solution clearly separates placement, frequency, security and cost considerations.

Constraint Application Summary

Constraint	Design Decision	Expected Benefit
Constraint	Design Decision	Expected Benefit
Limited APs	Capacity-based placement	APs are used where demand is highest.
High user count	5 GHz/6 GHz prioritized	Improves simultaneous-user capacity.
Interference	Channel planning + transmit-power control	Reduces co-channel interference.
Security	WPA2/WPA3-Enterprise + 802.1X/RADIUS	Provides individual authentication.
Segmentation	Student/staff/admin/guest VLANs	Limits unnecessary internal access.

Recommended Wireless Architecture

Internet/Core Network → Wireless Controller/Cloud Management → PoE Managed Switches → High-Density APs. Student, staff, administration and guest networks are separated using VLANs.

Question 3

A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Rubric-Based Answer

Understanding of Problem & Constraints ()

The institution handles sensitive customer information, requires controlled employee access, must comply with strict policies, should maintain network performance and must control operational cost.

Problem Decomposition ()

The problem is divided into segmentation, Internet perimeter protection, internal firewalling, authentication, IDS/IPS, monitoring, performance and cost.

Application of Constraints in Solution ()

Use VLAN/VRF segmentation for users, servers, databases, management and guests; use a DMZ for public services; deploy perimeter and internal segmentation firewalls; use MFA and 802.1X; deploy IDS/IPS and centralized logging.

Logical Reasoning & Strategy ()

Separating high-value databases from normal users limits lateral movement. A DMZ prevents public-facing services from being directly placed on internal networks. MFA reduces account compromise risk. Centralized monitoring improves detection without deploying many separate systems.

Solution Accuracy & Feasibility ()

The design is practical because segmentation can use managed switches and policy-based firewalls. Security inspection can be focused on sensitive zones to reduce unnecessary performance overhead.

Creativity & Innovation ()

Combining VLAN/VRF segmentation with an internal segmentation firewall provides layered security while controlling the number of security appliances.

Clarity of Presentation ()

The architecture is represented as logical zones with clear access policies and a constraint-based justification.

Constraint Application Summary

Constraint	Design Decision	Expected Benefit
Constraint	Design Decision	Expected Benefit
Security	VLAN/VRF + internal firewall	Limits lateral movement.
Performance	Selective inspection + QoS	Reduces unnecessary processing.
Compliance	MFA + logging + least privilege	Improves accountability and access control.
Budget	Centralized management/monitoring	Avoids unnecessary duplicated systems.
Scalability	Modular VLANs and policies	Allows new departments and services.

Recommended Secure Architecture

Internet → Perimeter Firewall → DMZ/Internal Core → Internal Segmentation Firewall → User/Server/Database Zones. IDS/IPS and centralized monitoring observe security events; management traffic uses a restricted management network.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand